



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2025

PH1006-1 – SEMICONDUCTOR PHYSICS AND PHOTONICS
(For ECE, EEE, VLSI & ACT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Physical Constants: Velocity of light, $c=3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h=6.63 \times 10^{-34} \text{ Js}$,
Electron mass, $m=9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e=1.6 \times 10^{-19} \text{ C}$,
Boltzmann constant, $k=1.38 \times 10^{-23} \text{ J/K}$. Avogadro number, $N_A = 6.022 \times 10^{26} / \text{kg mole}$.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **A wave packet is used to represent**
A) A light wave
B) A stationary wave
C) Matter wave
D) A transverse wave
2. **The concept of matter wave was suggested by**
A) Heisenberg
B) de Broglie
C) Schrodinger
D) Laplace
3. **The wavelength of matter waves does not depend on**
A) charge
B) mass
C) velocity
D) momentum
4. **The kinetic energy of electron and proton is the same. The relation between their de-Broglie wavelengths λ_e and λ_p is**
A) $\lambda_e = \lambda_p$
B) $\lambda_e < \lambda_p$
C) $\lambda_e > \lambda_p$
D) (a) $\lambda_e = 2\lambda_p$
5. **The square of the magnitude of the wave function is called**
A) current density
B) volume density
C) zero density
D) probability density
6. **The wave function of the particle lies in which region?**
A) $x > 0$
B) $x < 0$
C) $0 < x < L$
D) $x > L$
7. **Fermi level for a metal is**
A) Highest energy level occupied by electrons at 0°C
B) Highest energy level occupied by electrons at 0 K
C) Average value of all available energy levels
D) Addition of energy of all available electron energy levels
8. **Solids with high value of conductivity are called:**
A) Conductors
B) Semiconductor
C) Insulator
D) Non-metal
9. **A Silicon semiconductor has generally valence electrons.**
A) 2
B) 3
C) 6
D) 4
10. **As the doping to a pure semiconductor increases, the bulk resistance of the semiconductor**
A) Remains the same
B) Increases
C) Decreases
D) Becomes zero

11. Which of the following is known as indirect band gap semiconductors?
 A) Germanium B) GaAs
 C) GaAsP D) GaN
12. The Hall effect is used to determine:
 A) The type of charge carriers in a material B) The speed of sound in a solid
 C) The energy levels in an atom D) The density of protons in a metal
13. A Semiconductor hastemperature coefficient of resistance
 A) Positive B) Negative
 C) Zero D) Infinite
14. The following material exhibits Ferroelectricity?
 A) Iron B) Platinum
 C) Hydrogen D) Rochelle salt
15. The electric susceptibility of a dielectric material measures:
 A) Its resistance to electric current. B) Its ability to become magnetized in an electric field.
 C) How much it polarizes in response to an applied electric field. D) Its capacity to conduct electricity without losses.
16. All piezoelectric crystals posses
 A) A structure with center of symmetry B) A structure with no center of symmetry
 C) Cubic structure D) Rhombohedral structure
17. Why is laser light monochromatic?
 A) The excited electrons are in a metastable state. B) The system is in a state of population inversion.
 C) The emitted photon and incident photon are of the same phase. D) Emission happens between two well defined energy states
18. The emission of photon without being aided by any external agency is called
 A) light amplification B) induced absorption
 C) stimulated emission D) spontaneous emission
19. Optical fiber works on the phenomenon of _____
 A) polarization B) total internal reflection
 C) diffraction D) refraction
20. Which fiber is preferred for long distance communication?
 A) Step index single mode fiber B) Graded index multimode fiber
 C) Step index multimode fiber D) Both single mode and multimode fibers.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are matter waves? Mention its characteristics.	4	L*1	1	1,2
	b) Derive Schrodinger's time independent one-dimensional wave equation for a particle of mass m with energy E.	8	L2	1	1,2
	c) Calculate the momentum of an electron and the de-Broglie wavelength associated with it, if its kinetic energy is 1.5 keV.	4	L3	1	1,2
2.	a) Explain the following terms: (i) Probability density (ii) Normalization of a wave function.	6	L2	1	1,2
	b) Obtain the expression for energy Eigen value and Eigen function for a particle in a one-dimensional potential well of infinite height.	6	L2	1	1,2
	c) An electron is trapped in a one-dimensional region of length 1Å. How much energy must be supplied to excite the electron from the ground level to the first excited state?	4	L3	1	1,2

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|-------|--|---|-------|---|-----|
| 3. a) | What are the drawbacks (failure) of classical free electron theory? | 4 | L1,L2 | 2 | 1,2 |
| b) | What is Fermi energy and Fermi factor? Explain the dependence of Fermi factor for different energy level and temperature conditions. | 8 | L1,L3 | 2 | 1,2 |
| c) | Find the temperature at which there is 1% probability that a state 0.5 eV above Fermi energy is occupied. | 4 | L3 | 2 | 1,2 |

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | What are extrinsic semiconductors? Describe the mechanisms of carrier generation in extrinsic semiconductors. | 6 | L1 | 3 | 1,2 |
| b) | Explain the effect of temperature on the Fermi level in extrinsic semiconductor. | 6 | L2 | 3 | 1,2 |
| c) | Calculate the resistivity of intrinsic germanium if the intrinsic carrier density is $2.5 \times 10^{19} \text{ m}^{-3}$ assuming electron and hole mobilities of 0.38 and $0.18 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ respectively. | 4 | L3 | 3 | 1,2 |
| 5. a) | Explain the principle, construction and working of photo diode. | 4 | L1 | 3 | 1,2 |
| b) | State Hall Effect? Explain how Hall field is set up in the sample and obtain an expression for the Hall voltage and Hall coefficient of an n-type semiconductor. | 8 | L2 | 4 | 1,2 |
| c) | Sample of silicon is doped with 10^{22} phosphorus atoms per unit volume. What would you expect to measure for its resistivity? What Hall voltage would you expect in a sample 100 μm thick when a current of 1 mA is passed perpendicular to a magnetic field of 0.1T | 4 | L3 | 4 | 1,2 |
| 6. a) | Write a note on ferroelectric materials. | 4 | L2 | 4 | 1,2 |
| b) | Explain the term internal field and derive an expression for the same in case of liquids and solids. | 8 | L2 | 4 | 1,2 |
| c) | An elemental solid dielectric material has polarizability $7 \times 10^{-40} \text{ Fm}^2$. Assuming the internal field to be Lorentz field, calculate the dielectric constant for the material if the material has $3 \times 10^{28} \text{ atoms/m}^3$. | 4 | L3 | 4 | 1,2 |

Unit – III

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|-------|--|---|-------|---|-----|
| 7. a) | What is a LASER? Explain its properties. | 4 | L2 | 5 | |
| b) | With neat labelled diagrams, describe the construction and working of a Nd:YAG laser. What are its disadvantages? | 8 | L1,L2 | 5 | 1,2 |
| c) | Find the ratio of population of the two energy states the transition between which emits photons of wavelength 694.3 nm. Assume the ambient temperature as 27 °C. | 4 | L3 | 5 | 1,2 |
| 8. a) | Describe an optical fiber with working principle. Write any two advantages of an optical fiber communication system over conventional communication system. | 4 | L2 | 5 | 1,2 |
| b) | Describe the different types of optical fiber along with the sketches of geometry, refractive index profile and ray propagation. | 8 | L2 | 5 | 1,2 |
| c) | Calculate the numerical aperture and the acceptance angle for an optical fiber whose core and cladding has refractive index of 1.45 and 1.40 respectively. The light is launched from air. | 4 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
